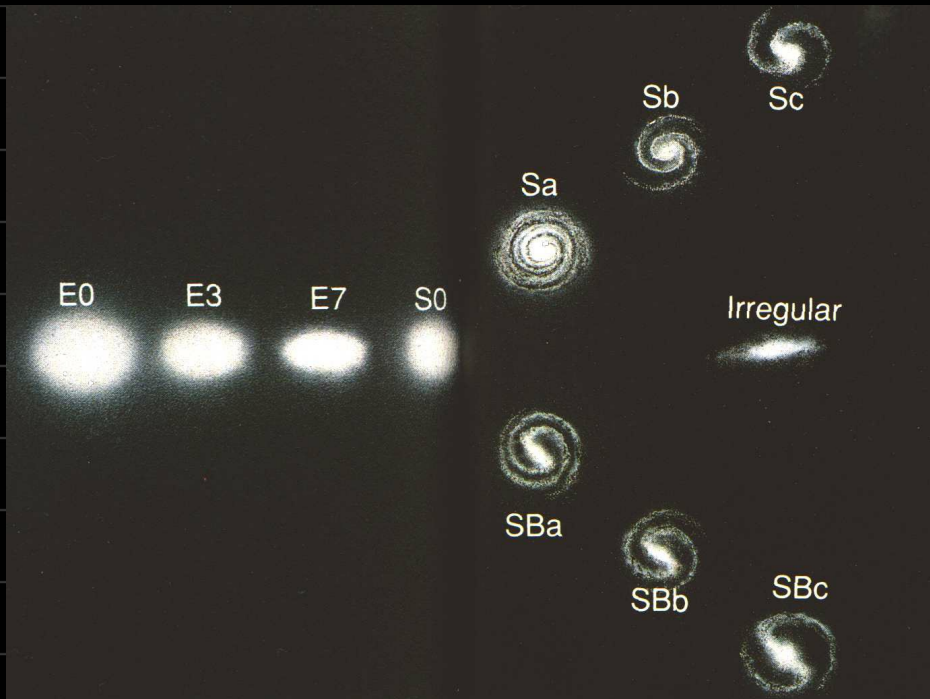


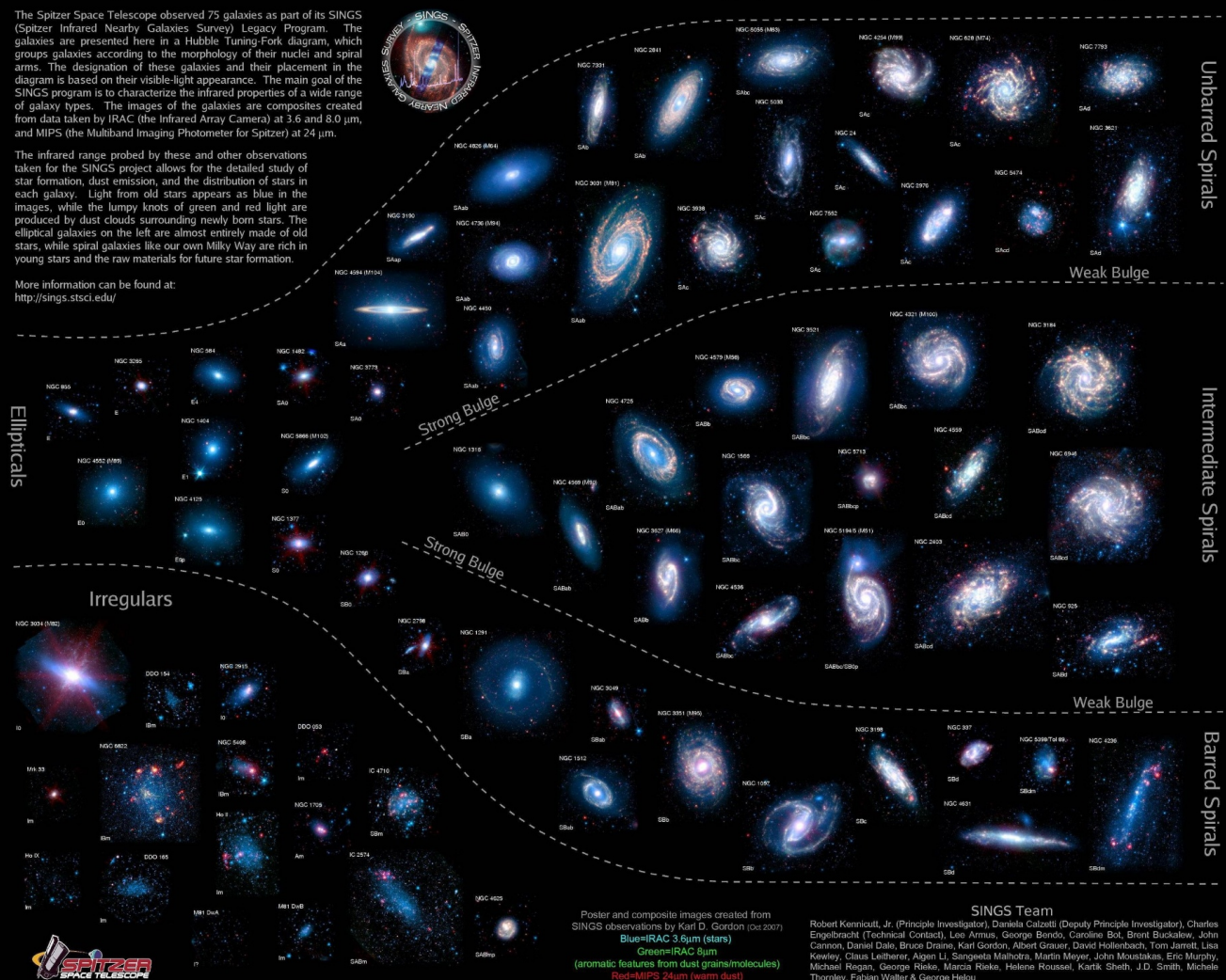


The Spitzer Infrared Nearby Galaxies Survey (SINGS) Hubble Tuning-Fork

The Spitzer Space Telescope observed 75 galaxies as part of its SINGS (Spitzer Infrared Nearby Galaxies Survey) Legacy Program. The galaxies are presented here in a Hubble Tuning-Fork diagram, which groups galaxies according to the morphology of their nuclei and spiral arms. The designation of these galaxies and their placement in the diagram is based on their visible-light appearance. The main goal of the SINGS program is to characterize the infrared properties of a wide range of galaxy types. The images of the galaxies are composites created from data taken by IRAC (the Infrared Array Camera) at 3.6 and 8.0 μm , and MIPS (the Multiband Imaging Photometer for Spitzer) at 24 μm .

The infrared range probed by these and other observations taken for the SINGS project allows for the detailed study of star formation, dust emission, and the distribution of stars in each galaxy. Light from old stars appears as blue in the images, while the lumpy knots of green and red light are produced by dust clouds surrounding newly born stars. The elliptical galaxies on the left are almost entirely made of old stars, while spiral galaxies like our own Milky Way are rich in young stars and the raw materials for future star formation.

More information can be found at:
<http://sings.stsci.edu/>



Poster and composite images created from SINGS observations by Karl D. Gordon (Oct 2007)
Blue=IRAC 3.6 μm (stars)
Green=IRAC 8.0 μm (aromatic features from dust grains/molecules)
Red=MIPS 24 μm (warm dust)

SINGS Team
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Few Notes:

ii) Inner region $M(r) \sim \sqrt{r}$ (solid)

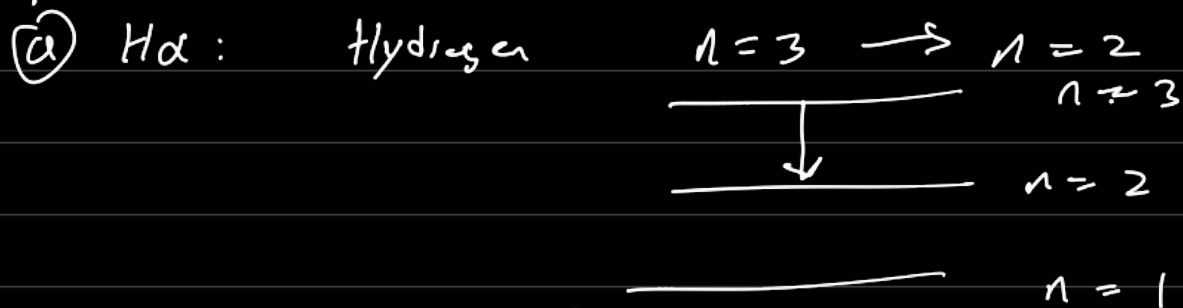
* $\rho(r) \sim \sqrt{r}$ (meanst)

$\rho \sim \frac{M}{V}$, $V \sim r^3 \Rightarrow r^{1/2} = \frac{M(r)}{r^3}$

$M(r) \sim r^{3.5}$

• Exp. values $M(r) = r^3 \sim r^{3.5}$

(ii) To determine velocities we use vsc



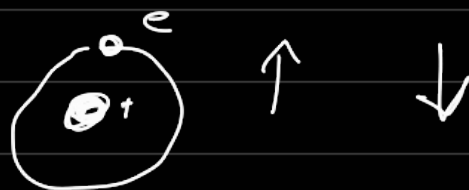
• Inner regions of the galaxy

(b) HI: 21cm line

• Spin flip of Hydrogen

$1/2 \rightarrow -1/2$

• Outer regions



* Normal Galaxies:

• Does not have an "active" center.

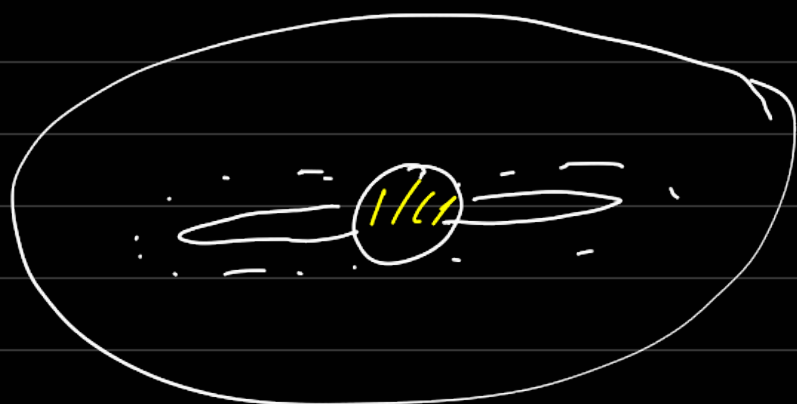
Hubble Classification:

(i) Spirals

(ii) Elliptical

(iii) Lenticular

(iv) Irregular



(i) Spirals:

• Central Bulge

• Spiral arms

→ Split into two

(a) Spiral-Barred (SB) $\sim 50\%$

(b) Spiral (S) $\sim 50\%$

* Have a decent idea as to how bars form but not 100% certain.

* Sub-sub categories: Sc, sb, b, c, cd, d

a $\rightarrow$ d: openness of spirals increases

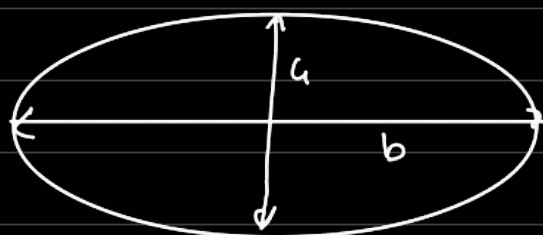
* MW: Sb

1B

- Classified by the ratio (Bulge/Disk) = $\left(\frac{b}{a}\right)$

Ellipticals: (Spheroidal) - pill shaped

- Do not have spirals
- Classified by ellipticity:



$$n = 10\left(1 - \frac{b}{a}\right)$$

- $a = b$: $n = 0$: circle

E_n : as n increases that means $b > a$

E_0 : circle

- Above: head on view:



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

- 3D picture

- Look different because their dynamics are different: they are not internally rotating: "Random motion"
- Smaller have the "total" rotation, larger do not
- Much more luminous than spirals

$$L \sim 100 L_{\text{MW}}$$

$$\Sigma(r) = \frac{\Sigma_0}{\left(1 + \frac{R}{a}\right)^2}$$

- Not much Dust
- Generally dominated by Red older stars
- Evolved a bit more than avg. spiral $\rightarrow$ "dynamically old"

Lenticular: Some in between E & S
S0, S0

- large disk + bulge

- Spiral structure faint $\rightarrow$ basically not exist
- Can also have bars

Irregular: Not symmetric, "lumpy"

- Low luminosity
- Blue $\rightarrow$ star forming regions

* Observing:

! depends on the Bands used

- B V, ...

Blue: young stars: clustered
dense looks sparse

Red: older: "better" idea but still incomplete

! Resolution: Spiral structure may not be resolved
 $\rightarrow$ Technology dependent

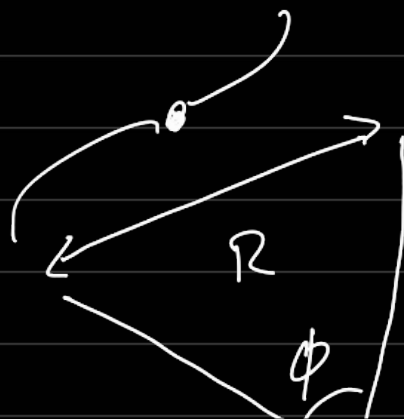
Understand their internal properties:

- $\rho, L, \sum L(r) dr,$
- Recall: looked at $(\frac{M}{L})$: mass-to-light ratio of the MW

$\rightarrow$ Bar vs an idea of probing radiating matter & Dark Matter

For MW: 7% was radiating
rest DM

$\rightarrow$ To do so: Surface Brightness
 $I(r, \phi)$



$I(r, \phi)$: Amount of light emitted per unit area on the sky

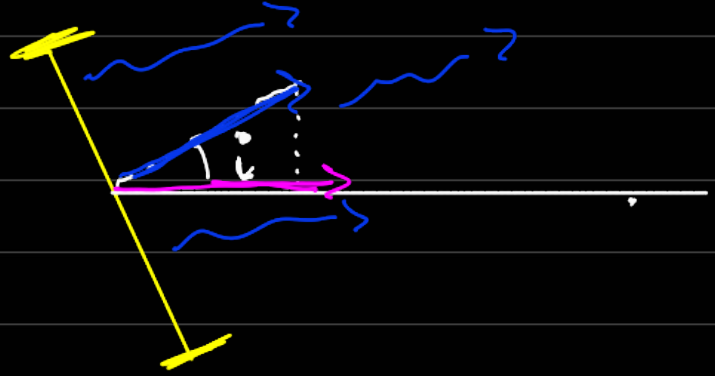
$$[I(r, \phi)] \sim \frac{\text{mag}}{[\text{arcsec}]^2}$$

• Usually average over ϕ :

$$I(r) = \langle I(r, \phi) \rangle \sim \int I(r, \phi) \tilde{\rho}(\phi) d\phi$$

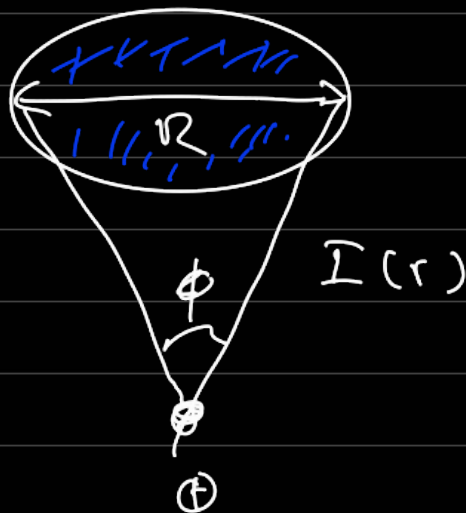
$I(r)$

$$L_{\text{obs}} = L \cos(i)$$



* Relate to Luminosity:

$$L = \int_0^{2\pi} d\phi \int_0^R dr I(r) r dr$$



* To actually find L : we need a surface brightness profile for galaxies:

$$I = \frac{L}{4\pi R^2} \quad \text{--- } R/R_h$$

For MW: $I(R) = I_0 e^{-R/R_h}$

R_h : some relative scale length

→ In general, for Spirals this profile holds

$$I(R) = I_0 e^{-R/R_h} \quad (\text{Spirals})$$

$$I(R) = \text{Flux} \sim \frac{L}{A}$$

$$L = \int_0^{2\pi} d\phi \int I(R) R dR = \underbrace{\int_0^{2\pi} d\phi}_{2\pi} \int_0^R I_0 e^{-R/R_h} R dR$$

$$L = 2\pi I_0 \int_0^R e^{-R/R_h} R dR \rightarrow R \rightarrow \infty$$

$$u \equiv R/R_h \Rightarrow du = \frac{dR}{R_h} \Rightarrow \underline{dR = R_h du}$$

$$L = 2\pi I_0 \int_0^R e^{-u} (u R_h) (R_h du)$$

$$= 2\pi I_0 R_h^2 \int_0^{R/R_h} e^{-u} u du$$

$$\int u e^{-u} du = \underline{1}$$

$$\textcircled{1} \quad L = 2\pi I_0 R_h^2 \quad \textcircled{2} \quad \text{total luminosity output}$$

$$\rightarrow \underline{L_{obs} = L \cos(i) = 2\pi I_0 R_h^2 \cos(i)}$$

